

Supplementary Material: Appendices for “Thermodynamic Approach and Saddle-Dominated Capacity of the LSE Dense Associative Memory”

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A Appendix A: Derivation of the prefactor in the non-retrieval sum

The Debye uniform asymptotic of the modified Bessel function (Olver et al. 2010, 10.41.3) gives the prefactor of Eq. (13). Keeping the leading ($k = 0$) term of the asymptotic series,

$$I_\nu(\nu z') \sim \frac{e^{\nu\eta(z')}}{(2\pi\nu)^{1/2}(1+z'^2)^{1/4}}, \quad (36)$$

where $z' = 2\beta N/(N-2)$ and

$$\eta(z') = \sqrt{1+z'^2} + \ln \frac{z'}{1+\sqrt{1+z'^2}}. \quad (37)$$

$\eta(z')$ is expanded to $\mathcal{O}(N^{-1})$ because it is multiplied by $\nu = N/2 - 1$ in the exponent:

$$\eta(z') = \eta(2\beta) + \eta'(2\beta) \frac{4\beta}{N} + \mathcal{O}(N^{-2}), \quad (38)$$

where

$$\eta(2\beta) = 1 + 2\beta\varphi^* + \ln \varphi^*, \quad (39)$$

$$\eta'(2\beta) = \varphi^* + \frac{1}{2\beta}. \quad (40)$$

Then

$$\nu\eta(z') = \frac{N}{2}\eta(2\beta) - \ln \varphi^* + \mathcal{O}(1/N). \quad (41)$$

The $\mathcal{O}(1)$ piece $-\ln \varphi^*$ in the exponent contributes a factor $1/\varphi^*$ to the prefactor. Using Stirling's formula

$$\Gamma(N/2) \sim \sqrt{4\pi/N} (N/2)^{N/2} e^{-N/2} \quad (42)$$

and collecting all terms in Eq. (18) to $\mathcal{O}(1)$ in the exponent and to leading order in the prefactor reproduces the exponent of Eq. (13) and gives the prefactor:

$$S_{\text{nr}} \approx \frac{e^{N[\alpha + g_{\text{max}}(\beta) - \beta]}}{\sqrt{1 - \varphi^{*4}}}, \quad (43)$$

with $g_{\text{max}}(\beta)$ given by Eq. (12). At $\beta = 1$, $\varphi^* \approx 0.618$ and the prefactor is ≈ 1.082 , reproducing Eq. (14).

B Appendix B: Tangency of the cusp lines to the spherical extreme

To show that the cusp lines (16) are tangent to the spherical-extreme curve (6) at every β , we need to show that the slope and the φ -intercept coincide.

Differentiating (6) once gives

$$\varphi'_{\text{max}}(\alpha) = \frac{1 - \varphi_{\text{max}}^2}{\varphi_{\text{max}}}, \quad (44)$$

and a second differentiation gives $\varphi''_{\text{max}}(\alpha) < 0$ for $\alpha > 0$, so φ_{max} is strictly concave on $(0, \infty)$. For a concave function every tangent line lies above the function, with equality only at the tangency point. This is used in the lower-envelope identity below.

The cusp line at sharpness β ,

$$\varphi = \alpha/\beta + g_{\text{max}}(\beta)/\beta, \quad (45)$$

has slope $1/\beta$ and φ -intercept $g_{\text{max}}(\beta)/\beta$. To find the point on the extreme curve where the tangent has the same slope, set the right-hand side of (44) equal to $1/\beta$:

$$\frac{1 - \varphi^2}{\varphi} = \frac{1}{\beta} \iff \beta \varphi^2 + \varphi - \beta = 0. \quad (46)$$

This is the saddle equation and its unique positive root is the spherical saddle (11). The corresponding α on the extreme curve follows from $\varphi^* = \varphi_{\text{max}}(\alpha^*)$:

$$\alpha^*(\beta) = -\frac{1}{2} \ln(1 - \varphi^{*2}). \quad (47)$$

A line of slope $1/\beta$ passing through $(\alpha^*(\beta), \varphi^*(\beta))$ has φ -intercept

$$b_{\text{tg}}(\beta) = \varphi^* - \alpha^*/\beta. \quad (48)$$

The cusp line has φ -intercept $g_{\text{max}}(\beta)/\beta$. Using (12), the φ -intercepts coincide; the two lines are therefore identical.

By the concavity statement above, the cusp line at any sharpness β lies above the extreme curve, with equality only at $\alpha = \alpha^*(\beta)$:

$$\varphi_c(\alpha, \beta) \geq \varphi_{\text{max}}(\alpha) \quad \text{for all } \alpha \text{ and all } \beta > 0, \quad (49)$$

with equality at one α for each β . Taking the infimum over β ,

$$\varphi_{\text{max}}(\alpha) = \inf_{\beta > 0} \varphi_c(\alpha, \beta), \quad (50)$$

the lower-envelope identity.

The saddle function $\varphi^*(\beta)$ (11) is continuous and strictly increasing on $(0, \infty)$ from 0 to 1. As β sweeps $(0, \infty)$, the tangency point $(\alpha^*(\beta), \varphi^*(\beta))$ therefore traces the entire extreme curve $\varphi_{\max}(\alpha)$, from $(0, 0)$ at $\beta \rightarrow 0$ to $\varphi_{\max} \rightarrow 1$ at $\beta \rightarrow \infty$.

Two results follow from the geometric setup above. At $T = 0$ the cusp line of slope $1/\beta$ reaches $\varphi = 1$ at $\alpha = \beta - g_{\max}(\beta)$, which is the zero-temperature capacity (17) (equivalently, the spinodal at $T = 0$).

The condition of Ramsauer et al. (2021), $\varphi_{\max}(\alpha) = 1 - \alpha/\beta$, is the intersection of $\varphi_{\max}$ with the line of slope $-1/\beta$ through $(0, 1)$. By (50), $\varphi_{\max}$ lies below every cusp line, so this line crosses $\varphi_{\max}$ at strictly smaller α than the cusp line at slope $1/\beta$ reaches $\varphi = 1$. Their capacity α_c^R therefore satisfies $\alpha_c^R < \alpha_c(\beta, 0)$ for every $\beta > 0$.

Using (11) in (12) and then in (17), one obtains:

$$\begin{aligned} \alpha_c(\beta, 0) &= \beta - g_{\max}(\beta) \rightarrow \beta & (\beta \rightarrow 0), \\ \alpha_c(\beta, 0) &= \beta - g_{\max}(\beta) \rightarrow \frac{1}{2} \ln \beta + \frac{1}{2} & (\beta \rightarrow \infty), \end{aligned} \quad (51)$$

the small- and large- β limits of the zero-temperature capacity cited in the main text.

Legendre transform. The lower-envelope identity (50) and the formula below are the two sides of a Legendre transform between $\varphi_{\max}(\alpha)$ and $g_{\max}(\beta)/\beta$. The forward transform takes $\varphi_{\max}$ to its conjugate,

$$g_{\max}(\beta)/\beta = \sup_{\alpha} [\varphi_{\max}(\alpha) - \alpha/\beta], \quad (52)$$

with the supremum attained at $\alpha = \alpha^*(\beta)$ from the slope-match condition (46). The Legendre transform is involutive: applying it to $g_{\max}(\beta)/\beta$ recovers $\varphi_{\max}(\alpha)$, which is exactly (50) after using the cusp-line definition (16).

Differentiating (12) in β and using the envelope theorem $\partial g / \partial \varphi|_{\varphi^*} = 0$ gives

$$g'_{\max}(\beta) = \varphi^*(\beta), \quad (53)$$

so $g_{\max}(\beta)$ is recovered by integrating $\varphi^*(\beta)$. Plugging the small- and large- β limits $\varphi^* \sim \beta$ and $\varphi^* \sim 1 - 1/(2\beta)$ into (53) and integrating reproduces (51).

References

- Olver, F. W. J.; Lozier, D. W.; Boisvert, R. F.; and Clark, C. W., eds. 2010. *NIST Handbook of Mathematical Functions*. New York: Cambridge University Press. ISBN 978-0-521-19225-5.
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